JUNCTION IQ - SMART TRAFFIC MANAGEMENT SYSTEM FOR URBAN CONGESTION CONTROL

A MINI PROJECT 3 REPORT

Submitted by

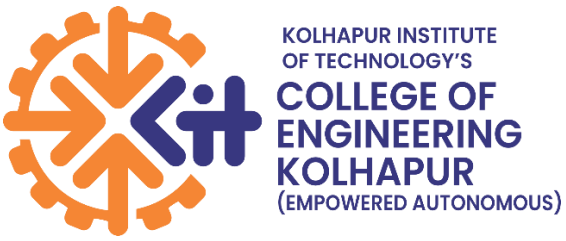
Anusha Patil (2425000174) (B-44)
Shrutika Patil (2425001031) (B-50)
Adarsh Gurav (2425000776) (B-73)
Sanket Magdum (2425000540) (B-77)

in partial fulfillment for the award of the degree of

THIRD YEAR B-TECH

IN

COMPUTER SCIENCE ENGINEERING



KOLHAPUR INSTITUTE OF TECHNOLOGY'S

COLLEGE OF ENGINEERING KOLHAPUR

(EMPOWERED AUTONOMOUS)

OCT 2026

KOLHAPUR INSTITUTE OF TECHNOLOGY'S COLLEGE OF ENGINEERING KOLHAPUR (EMPOWERED AUTONOMOUS)

CERTIFICATE

This is to certify that the Mini Project 3 report entitled, **"Junction IQ- Smart Traffic Management System for Urban Congestion Control."** submitted by **"Anusha Patil (B-44), Shrutika Patil (B-50), Adarsh Gurav (B-73), Sanket Magdum (B-77)"** , in partial fulfillment for the award of the degree of **" THIRD YEAR B-TECH"** in **"COMPUTER SCIENCE AND ENGINEERING"** at Kolhapur Institute of Technology's College of Engineering Kolhapur (Empowered Autonomous), Kolhapur, Maharashtra, INDIA, is a record of his / her own work carried out under my / our supervision and guidance.

SIGNATURE	SIGNATURE	SIGNATURE
Dr. Lingaraj Hadimani	Mrs.Yogita Narule	Mr. Sanjay Patil
HEAD OF THE DEPARTMENT	**SUPERVISOR**	**PBL EVALUATOR**

KOLHAPUR INSTITUTE OF TECHNOLOGY'S COLLEGE OF ENGINEERING KOLHAPUR (EMPOWERED AUTONOMOUS)

DECLARATION

We hereby declare that the **Mini Project III** entitled, **Junction IQ - Smart Traffic Management System for Urban Congestion Control.** submitted to Kolhapur Institute of Technology's College of Engineering Kolhapur (Empowered Autonomous), Kolhapur, Maharashtra, INDIA in the partial fulfillment of the award of the Degree of **"THIRD YEAR B-TECH"** in **"COMPUTER SCIENCE AND ENGINEERING"** is a bonafide work carried out by us. The material contained in this Project has not been submitted to any University or Institution for the award of any degree.

NAME OF THE STUDENT(S)

Anusha Patil (2425000174) (B-44)
Shrutika Patil (2425001031) (B-50)
Adarsh Gurav (2425000776) (B-73)
Sanket Magdum (2425000540) (B-77)

Place: Kolhapur
Date:

KOLHAPUR INSTITUTE OF TECHNOLOGY'S COLLEGE OF ENGINEERING KOLHAPUR (EMPOWERED AUTONOMOUS)

ACKNOWLEDGEMENT

We take this opportunity to express our sincere gratitude to all those who supported and guided us in the successful completion of our project titled "**Junction IQ-Smart Traffic Management System for Urban Congestion Control.**" We are deeply thankful to our respected guide **Mrs. Yogita Narule** and PBL Evaluator **Mr. Sanjay Patil** for valuable guidance, suggestions, and constant encouragement throughout the development of this project. We also express our heartfelt thanks to our Head of Department, **Dr. Lingaraj Hadimani**, for providing the required facilities and a motivating environment for our work. Our sincere appreciation extends to all the faculty members of the Department of Computer Science And Engineering for their helpful feedback and assistance. Finally, we thank our friends and family for their continuous support and motivation, which greatly contributed to the successful completion of this project.

NAME OF THE STUDENT(S)

Anusha Patil (2425000174) (B-44)
Shrutika Patil (2425001031) (B-50)
Adarsh Gurav (2425000776) (B-73)
Sanket Magdum (2425000540) (B-77)

Place: Kolhapur
Date:

KOLHAPUR INSTITUTE OF TECHNOLOGY'S COLLEGE OF ENGINEERING KOLHAPUR (EMPOWERED AUTONOMOUS)

ABSTRACT

Junction IQ - Smart Traffic Management System for Urban Congestion Control

Junction IQ - The Smart Traffic Management System for Urban Congestion Control aims to reduce traffic congestion, minimize vehicle waiting time, and improve traffic flow at intersections. Traditional traffic signals use fixed timings that may not match actual traffic conditions, resulting in unnecessary delays and uneven congestion.

The proposed system uses **SUMO, NetEdit, Python, and TraCI API** to simulate traffic and dynamically adjust signal timings based on vehicle count, lane density, vehicle speed, and waiting time. A rule-based decision-making mechanism determines green-light durations and provides priority to ambulances according to predefined rules.

The system's performance is evaluated by comparing dynamic signal control with conventional fixed-time control under simulated traffic conditions. The recorded results show a reduction in average vehicle waiting time from **118 seconds under fixed-time control to 70.48 seconds under dynamic-time control, representing an improvement of approximately 40.27%**. These results demonstrate the potential of traffic-responsive signal control to reduce unnecessary waiting and improve emergency vehicle movement within a simulated road network. The proposed system can be further extended to support complex road networks, coordinated traffic signals at multiple intersections, and potential real-world deployment.

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Chapter 1

INTRODUCTION

1.1 Project Category

Our project, **Junction IQ - Smart Traffic Management System for Urban Congestion Control**, belongs to the domain of Intelligent Transportation Systems (ITS) and Smart City Solutions. It focuses on dynamically controlling traffic signals based on vehicle count and traffic conditions while providing priority to emergency vehicles such as ambulances. The system uses **SUMO** for traffic simulation, **NetEdit** for designing road networks and intersections, **Python** for implementing traffic control logic, and the **TraCI API** to establish communication between Python and SUMO. These technologies enable the system to simulate different traffic scenarios, adjust signal timings, and evaluate its effectiveness in reducing congestion and vehicle waiting time.

1.2 Problem Formulation

Traditional traffic signals generally operate with fixed timings that do not adapt to changing traffic conditions, resulting in long queues, uneven traffic distribution across lanes, and unnecessary vehicle waiting time. During peak hours, some lanes experience heavy congestion while others remain relatively less occupied, leading to inefficient use of green-light time. Additionally, emergency vehicles such as ambulances may face delays at intersections because conventional traffic signals do not always provide priority during emergencies. The absence of automatic, traffic-responsive signal control further contributes to congestion and increased travel time. The objective of this project is to develop a Smart Traffic Management System that dynamically adjusts signal timings based on vehicle count and lane density while providing priority to emergency vehicles through predefined rules. The proposed system aims to reduce vehicle waiting time, improve traffic flow, minimize congestion, and evaluate the effectiveness of dynamic signal control through SUMO simulation**.**

1.3 Identification / Reorganization of Need

The need for this project arises from the limitations of fixed-time traffic signals in managing changing traffic conditions. During peak hours, some lanes experience heavy congestion, while others have fewer vehicles, resulting in inefficient signal timing and increased waiting time. Therefore, an intelligent system is required to analyze traffic conditions and allocate green-light duration according to actual traffic demand. The major requirements include **dynamic control** to adjust signal timings based on congestion, **traffic data collection** to analyze vehicle count, lane density, speed, and waiting time, and **rule-based decision-making** to determine suitable green-light durations within predefined limits. The system also requires an **emergency vehicle prioritization mechanism** to provide ambulances with priority under specified conditions. Finally, **performance evaluation** is carried out using SUMO simulation by comparing dynamic-time and fixed-time signal control to assess improvements in traffic flow and vehicle waiting time.

1.4 Existing System

The existing traffic management systems discussed in the literature review primarily use fixed-time signals, adaptive traffic control algorithms, and simulation-based approaches to manage traffic congestion. Several studies have explored fuzzy logic, rule-based decision-making, and artificial intelligence techniques to adjust signal timings according to traffic conditions. SUMO has been widely used to simulate traffic scenarios and evaluate improvements in vehicle waiting time, traffic flow, and congestion. Some research studies also focus on emergency vehicle prioritization using GPS, RFID, IoT sensors, and dynamic signal control. However, these approaches have certain limitations, including dependence on costly infrastructure, limited real-world implementation, and difficulties in handling complex traffic conditions. Some studies focus on individual intersections or road network optimization rather than integrating dynamic signal timing and emergency vehicle priority into a single system. The existing research provides a foundation for developing intelligent traffic management solutions, but there is still scope for a simulation-based system that combines vehicle count analysis, rule-based dynamic signal timing, and ambulance prioritization. Our proposed project addresses this need by using **SUMO, Python, and the TraCI API** to adjust traffic signal timings according to simulated traffic conditions and evaluate system performance by comparing fixed-time and dynamic-time signal control.

Name	Year	Purpose of the Study	Methodology	Limitations	Relevance to Our Work
Adaptive Traffic Signaling Control Using SUMO Simulator	International Journal of Innovative Science and Research Technology 2025	To develop a two-stage, three-module **fuzzy logic** system for real-time management of signalized junctions to reduce average vehicle waiting times.	Implemented a fuzzy logic traffic management system using the **SUMO simulation** tool with real-world traffic data from a congested intersection in Kilis, Turkey. The system dynamically adjusts signal timing using 52 fuzzy rules and was compared against conventional fixed-time control.	The study concentrates exclusively on an isolated intersection rather than a multi-intersection network.	Proves dynamic signal control in SUMO outperforms fixed-time systems. Uses real-time vehicle counts and queue lengths to adjust timings.
A SUMO Based Simulation Framework for Intelligent Traffic Management System	Journal of Traffic and Logistics Engineering Vol. 8, No. 1 2020	To explore a **Deep-Neuro-Fuzzy model** within **the SUMO traffic** simulation environment and develop a GUI to control simulation input attributes for optimal route planning.	Utilized C-means clustering and a Deep-Neuro-Fuzzy framework to generate dynamic road weights based on environmental and artificial attributes. Applied routing algorithms (Dijkstra, A*, CHWrapper) in SUMO to find optimal vehicle paths.	The model has not yet been implemented in real-time physical road networks, which would require costly personalized infrastructure. Other hidden parameters of existing SUMO algorithms remained unchanged during testing.	Shows how to integrate SUMO with an external backend to update traffic logic. Validates extracting live metrics and sending custom commands back to SUMO
AI-Powered Traffic Signal Control: A Deep Reinforcement Learning Framework for Urban Congestion Reduction	International Journal of Environmental Sciences 2025	To explore AI-based traffic signal regulation using deep **reinforcement learning** to minimize vehicle waiting times, maximize flow efficiency, and enhance pedestrian and emergency vehicle safety..	Formulated traffic signal optimization as a sequential decision-making task using DRL techniques like Proximal Policy Optimization (PPO), Deep Q-Networks (DQN), and Advantage Actor-Critic (A3C). Performed simulations using **SUMO with real-world** and synthetic data to compare AI methods against fixed-timing controls.	Relied on simulation-based testing, which did not fully account for erratic driving habits, infrastructural limitations, or external weather factors	Highlights using SUMO to prioritize emergency vehicles and specific vehicle classes. Confirms the method of using live queue data to optimize traffic lights.

Table 1 . Literature Review

Name	Year	Purpose of the Study	Methodology	Limitations	Relevance to Our Work
Enhancing Traffic Monitoring Systems through Image Analysis and Rule-Based Machine Learning	International Conference on Machine Learning and Data Engineering 2025	To present an advanced traffic monitoring system integrating **reasoning-based image analysis and rule-based machine** learning to improve traffic management, reduce congestion, and enhance road safety.	Leverages visual data from spot sensors utilizing a three-tier module approach. It processes daytime scenes via Google cloud using the **RCNN algorithm** and nighttime scenes via bimodal histograms and Otsu's thresholding for headlight extraction. A high-level **module applies rule-based reasoning** (via an inference engine and knowledge base) to refine the models.	Real-world testing was restricted to favorable weather, meaning the system's robustness in environmental conditions like fog or snow remains untested.	It proves that adding rule-based reasoning helps systems instantly adapt to unpredictable, real-time traffic scenarios.
Development of the Rule Based Approach to Traffic Management	2017 (the paper references the 2017 methodology and evaluation work)	To develop a Rule-Based Traffic Management (RBTM) approach that can manage traffic dynamically, detect congestion, handle conflicts, and distribute traffic across the road network more efficiently.	The study uses business rules, decision tables, traffic data, road-priority rules, and multi-agent concepts. The system detects traffic conditions such as saturation, congestion and gridlock, then requests actions such as increasing outbound flow, limiting inbound flow, or rerouting traffic.	The approach can shift traffic problems to lower-priority roads, increasing waiting queues there. It also caused an increase in red-light runners, which was an undesirable result. The second pilot did not use rerouting because suitable signs were unavailable.	This research helps in detecting traffic congestion and managing traffic automatically. It provides ideas for rule-based decision-making and traffic control. These concepts can help us improve traffic flow and reduce congestion.
Traffic Intersection Simulation Using Turning Movement Count Data in SUMO: A Case Study of Toronto Intersections	2025 — the paper was posted on arXiv on 14 August 2025	To develop **CrossFlow**, a tool that uses real-world traffic data to create realistic intersection simulations in **SUMO**, reducing the need for synthetic or randomly generated traffic data.	Real-world Turning Movement Count (TMC) data from Toronto was converted into SUMO-compatible traffic flows. A Python-based CrossFlow GUI was used to create the network, generate vehicle flows, and run simulations.	The study mainly focuses on regular 3-way and 4-way intersections. Pedestrian and bicycle data were not included, and individual vehicle-level control is limited because the study uses aggregated flow data.	It helps in real-time traffic analysis and congestion detection using real traffic data. It also provides a simulation-based approach for testing traffic signal optimization and improving traffic flow.

Table 2. Literature Review

Name	Year	Purpose of the Study	Methodology	Limitations	Relevance to Our Work
Rule-Based Dynamic Traffic Management for Emergency Vehicle Routing: A Smart Infrastructure Approach	**2019** — Published in June 2019	To develop an intelligent traffic management system that gives **priority to emergency vehicles**, dynamically adjusts traffic signals, and optimizes routes using real-time traffic, IoV, weather, and road-network data.	The proposed **Rule-Based Dynamic Traffic Management (RBD-TM)** combines real-time data collection, IoV/IoT sensors, GPS, RFID, shortest-path algorithms such as **Dijkstra/Floyd-Warshall**, GNNs, and dynamic traffic-signal control.	The paper does not clearly provide a separate limitations section. However, the proposed system depends on **real-time sensors, IoV communication, GPS/RFID, and traffic infrastructure**, which are required for effective operation.	This research helps in **detecting traffic congestion and managing traffic dynamically**. It provides ideas for **emergency vehicle priority, automatic signal control, and route optimization**, which can help improve traffic flow and reduce response time.
Dynamic Emergency Vehicle Prioritization in a Connected Urban Environment: A Rule-Based Approach	Published in June 2024	To develop an **Emergency Vehicle Priority Management (EVPM)** system that dynamically adjusts traffic signals and vehicle lanes to give priority to emergency vehicles while minimizing disruption to normal traffic.	The system uses a **rule-based algorithm** with IoV, 5G, GPS, V2X, RFID and sensors. It detects and tracks emergency vehicles, clears their lane, and changes traffic signals to green when required. The system was tested using **SUMO** with real-world **TAPASCologne** traffic data.	The study identifies **high initial costs and data reliability** as challenges. It also notes that lane clearance may not work optimally during severe congestion.	This research helps in **dynamic traffic management and emergency vehicle prioritization**. It provides useful ideas for **automatic signal control, lane clearance, real-time traffic monitoring, and rule-based decision-making**.
Traffic Intersection Simulation Using Turning Movement Count Data in SUMO: A Case Study of Toronto Intersections	**2025** — posted on arXiv on 14 August 2025.	To develop **CrossFlow**, a tool that uses real-world Turning Movement Count (TMC) data to create realistic traffic simulations in **SUMO**, instead of using synthetic or randomly generated data.	Real-world traffic data from Toronto was converted into **SUMO-compatible vehicle flows**. A Python-based **CrossFlow GUI** was used to create the road network, generate traffic flows, and run simulations.	The study mainly focuses on **3-way and 4-way intersections**. Pedestrian and bicycle data were outside the study scope, and individual vehicle-level control is limited because aggregated flow data is used.	This research helps in **real-time traffic analysis and congestion management**. It provides useful ideas for **traffic simulation, signal optimization, and improving traffic flow** using real-world data.

Table 3. Literature Review

Name	Year	Purpose of the Study	Methodology	Limitations	Relevance to Our Work
AI-based traffic control and management systems – a synthesis of Macedonian research experiences	**2025 — published in *Transportation Research Procedia***	The study aims to explore the use of **AI and Machine Learning** in traffic management. It focuses on reducing traffic congestion and vehicle emissions while improving traffic flow and throughput.	The study uses **two case studies**. The first applies **Reinforcement Learning with Q-learning** for adaptive traffic signal control using VISSIM simulation, while the second uses **AI-based video processing** to collect and analyze traffic data.	The study mainly relies on **simulation-based testing**, so real-world implementation requires further validation. Some traffic video data was also incomplete because of unsuitable camera positioning.	This research is relevant because it provides useful approaches for **AI-based traffic monitoring and intelligent traffic control**. The use of Q-learning, video analysis, and real-time traffic data can provide a foundation for developing our proposed traffic-management system.
Computation and Optimization of Traffic Network Topologies Using Eclipse SUMO.	**The research paper was published in 2022 in *Annals of Emerging Technologies in Computing (AETiC)***	The study aims to analyze and optimize **urban traffic network topology** using traffic simulation. It specifically compares one-way and two-way street configurations to understand their impact on traffic congestion, travel time, traffic flow, and road capacity.	The researchers used **Eclipse SUMO**, a microscopic traffic simulation tool, to model traffic in Downtown Brickfields, Kuala Lumpur. They simulated one-way and two-way street networks and evaluated traffic speed, flow rate, density, and trip completion time under different traffic conditions.	The study is mainly based on **computer simulation** rather than real-time implementation. The results are specific to the selected road network and traffic scenarios, so different locations and traffic conditions may produce different results. The study also focuses mainly on street topology rather than complete intelligent traffic-control systems.	This research is relevant because it shows how **traffic simulation and network optimization** can be used to understand and reduce congestion. Eclipse SUMO, traffic-flow analysis, and comparison of different road configurations can provide a useful foundation for our proposed intelligent traffic-management system.

Table 4. Literature Review

1.5 Proposed System

The proposed **Junction IQ - Smart Traffic Management System for Urban Congestion Control** uses SUMO, Python, and the TraCI API to manage traffic signals dynamically. The system analyzes vehicle count, lane density, vehicle speed, and waiting time to determine suitable green-light durations based on traffic demand. A rule-based mechanism adjusts signal timings within predefined limits and prioritizes ambulances under specific conditions. The system is tested in SUMO under different traffic scenarios, including varying congestion levels and emergency situations. Its performance is evaluated by comparing fixed-time and dynamic-time signal control using parameters such as vehicle waiting time, queue length, and traffic flow. The main aim is to reduce traffic congestion, minimize delays, and improve emergency vehicle movement.

1.6 Objectives

The main objective of the **Junction IQ - Smart Traffic Management System for Urban Congestion Control** are :

- To Build Traffic Simulation
- To Develop a rule based logic model to manage congestion
- To Develop Smart Traffic Signals by dynamic timer
- To Give Priority to Ambulances
- To Evaluate System Performance

1. 7 Unique Features

The system incorporates a **rule-based decision engine** that analyzes traffic conditions and applies predefined rules to determine suitable green-light durations while maintaining signal timing limits. **SUMO and TraCI** are used for simulation-based testing under different traffic scenarios, including varying vehicle types, congestion levels, and emergency vehicle situations. The system's performance is evaluated by comparing fixed-time and dynamic-time traffic signal control. The recorded results show a reduction in average waiting time from 118 seconds to 70.48 seconds, representing a **40.27 % reduction** under the tested simulation conditions.

Chapter 2

REQUIREMENT ANALYSIS & SYSTEM SPECIFICATION

2.1 Feasibility Study

The **Smart Traffic Management System for Urban Congestion Control** is evaluated based on technical, economic, and operational feasibility to ensure its practicality and effectiveness. **Technical Feasibility:** The project uses SUMO, NetEdit, Python, and the TraCI API for road network design, traffic simulation, and dynamic traffic signal control. The system adjusts signal timings based on vehicle counts and traffic congestion while providing priority to emergency vehicles such as ambulances and fire brigades. **Economic Feasibility:** The project is cost-effective because all the required software tools are open-source and freely available, reducing development and licensing expenses. It can be developed and tested on an existing computer without requiring expensive hardware or physical traffic infrastructure. **Operational Feasibility:** The system can be tested in a simulated environment under different traffic conditions, including normal traffic, heavy congestion, and emergency situations. Its performance can be evaluated using vehicle waiting time, queue length, traffic flow, and emergency vehicle waiting time.

2.2 Software Requirement Specification (SRS)

The Software Requirement Specification defines the hardware and software required to develop and execute the project. **Software Requirements:** SUMO is used for traffic simulation, NetEdit for road and intersection design, Python for implementing the rule-based decision engine, and the TraCI API for communication between Python and SUMO. **Hardware Requirements:** A computer or laptop capable of running SUMO simulations and Python programs is required. Sufficient RAM and processing power help execute simulations smoothly. **Functional Requirements:** The system must collect vehicle count, lane density, speed, and waiting time, dynamically determine green-light durations, and provide ambulance priority according to predefined rules.

2.3 Validation

Validation ensures that the proposed system works according to its objectives and produces the expected results.The system is tested using different traffic scenarios in SUMO to verify vehicle data collection, dynamic signal timing, and emergency vehicle prioritization. The results of dynamic-time control are compared with fixed-time control using average waiting time as a key performance parameter. In the recorded simulation results, average waiting time decreased from **118 seconds to 70.48 seconds**, representing a **40.27% reduction**. This indicates improved performance under the tested simulation conditions. Additional testing is required to evaluate the system under different traffic densities and emergency situations.

2.4 Expected Hurdles

During development, the project may face challenges in connecting Python with SUMO through the TraCI API and ensuring accurate collection of vehicle information. Incorrect vehicle counts or traffic data may lead to unsuitable signal timing decisions.Another challenge is implementing emergency vehicle prioritization without creating unnecessary delays for other lanes. Designing appropriate green-light durations and handling changing traffic conditions may also require repeated testing.These hurdles can be addressed through systematic debugging, testing individual modules, verifying vehicle information, and evaluating the system under different traffic scenarios.

2.5 SDLC Model

The **Iterative SDLC Model** is suitable for this project because the traffic management system can be developed, tested, and improved through repeated cycles. Initially, project requirements are identified and the road network is designed using NetEdit. Next, traffic simulation is configured in SUMO, and Python is connected through TraCI to implement the rule-based signal control mechanism. The system is then tested for vehicle count-based signal timing, traffic congestion, and emergency vehicle prioritization. The results are analyzed, and necessary modifications are made to improve system performance. This process continues until the system achieves its intended objectives.The iterative approach allows errors to be identified early and enables the traffic control logic to be improved gradually.

Chapter 3

SYSTEM DESIGN

3.1 Design Approach

The Junction IQ - Smart Traffic Management System for Urban Congestion Control follows a modular design approach in which the complete system is divided into smaller modules, each performing a specific task. The main modules include road network design, traffic simulation, vehicle data collection, traffic condition analysis, dynamic signal timing, and emergency vehicle prioritization. NetEdit is used to design roads, lanes, and intersections, while SUMO simulates vehicle movement under different traffic conditions. Python processes the traffic information and applies predefined rules to determine suitable signal timings. The TraCI API establishes communication between Python and SUMO, allowing the system to collect traffic data and update signal states during simulation. This approach makes the system easier to develop, test, debug, and improve.

3.2 Detailed Design

The detailed design describes the internal working of each module and how the modules interact with one another. First, a road network with multiple incoming lanes and traffic signals is created using NetEdit. SUMO is then configured to simulate different types of vehicles, such as cars, bikes, trucks, and ambulances. During simulation, Python uses the TraCI API to collect vehicle count, lane density, vehicle speed, and waiting time. These parameters are analyzed by a rule-based decision engine to determine the green-light duration for each lane. Lanes with greater traffic demand can receive higher green-time priority, while predefined limits prevent signal durations from becoming unsuitable. The system also checks for emergency vehicles and applies the configured priority rules when an ambulance is detected. Finally, Python sends the signal control commands to SUMO, and the updated traffic conditions are observed through the simulation.

3.3 Structured Design Tools

The project uses SUMO, NetEdit, Python, and the TraCI API as its main design and development tools. SUMO provides a microscopic traffic simulation environment in which individual vehicles and their movements can be observed. NetEdit is used to create and modify road networks, lanes, junctions, and traffic signal configurations. Python is responsible for implementing the traffic analysis and rule-based decision-making logic.The TraCI API acts as a communication interface between Python and SUMO, enabling the collection of vehicle information and the control of traffic signals while the simulation is running. These tools work together to develop and evaluate the proposed dynamic traffic management system. Their modular integration also makes it easier to modify traffic rules and test different scenarios.

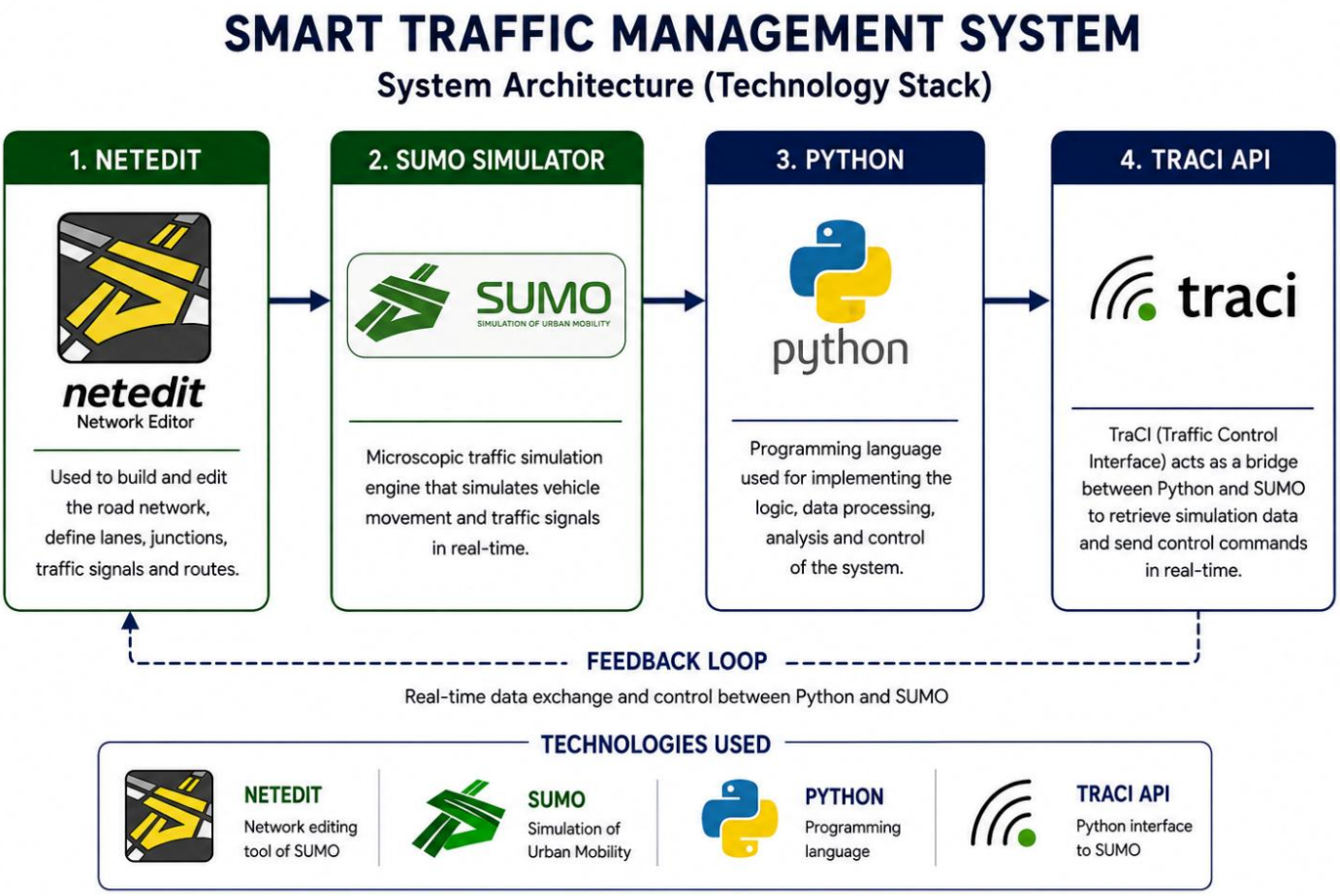


Diagram 1. System Architecture

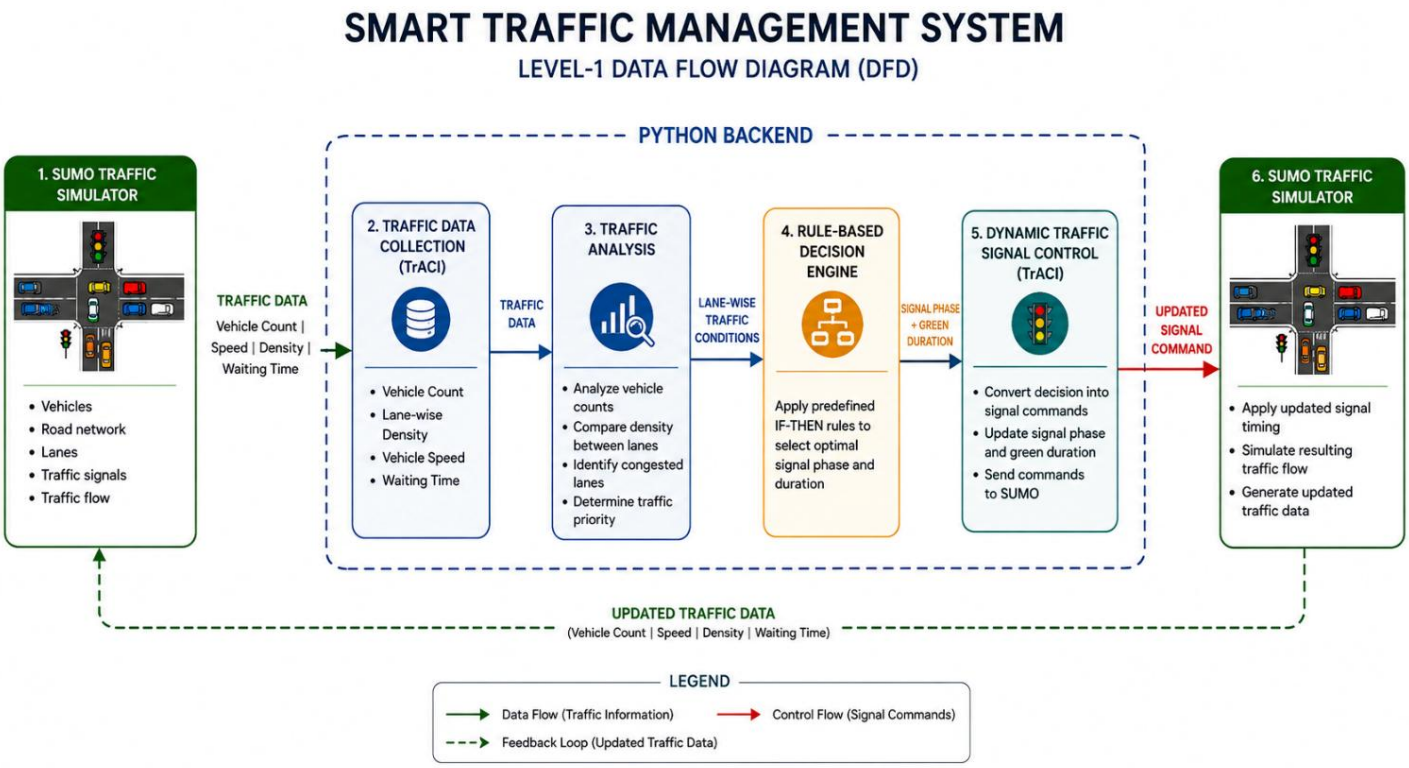


Diagram 2. Data Flow Diagram

3.4 User Interface Design

The user interface of the project is primarily based on the SUMO graphical user interface (GUI), which provides a visual representation of the simulated road network and traffic conditions. It displays roads, lanes, vehicles, and traffic signals, allowing users to observe the movement of vehicles through intersections.During simulation, users can monitor how traffic signals change according to the implemented control logic and how vehicles respond to different green-light durations. Different vehicle types, including emergency vehicles, can be observed to evaluate their movement through the road network. The interface also helps identify congestion, long vehicle queues, and delays under different traffic conditions.This visual representation makes it easier to understand system behaviour, test signal control decisions, and compare fixed-time and dynamic-time traffic management.

3.5 Methodology

The project methodology consists of road network creation, traffic simulation, data collection, traffic analysis, signal control, and performance evaluation. Initially, the road network and intersections are designed using NetEdit. The network is then configured in SUMO with appropriate vehicle routes, vehicle types, and traffic signals. Next, Python is connected to SUMO through the TraCI API to collect traffic information during simulation.The collected parameters, including vehicle count, lane density, speed, and waiting time, are analyzed by a rule-based decision engine. Based on the traffic conditions, the system determines suitable green-light durations and updates the traffic signals. If an ambulance is detected, the predefined emergency-priority mechanism is applied.The simulation continues to collect updated traffic information and adjust signal timings accordingly. Finally, the performance of dynamic-time control is compared with fixed-time control using average waiting time and other relevant traffic observations.

3.6 System Design Flowchart

Below is the logical flow of the system:

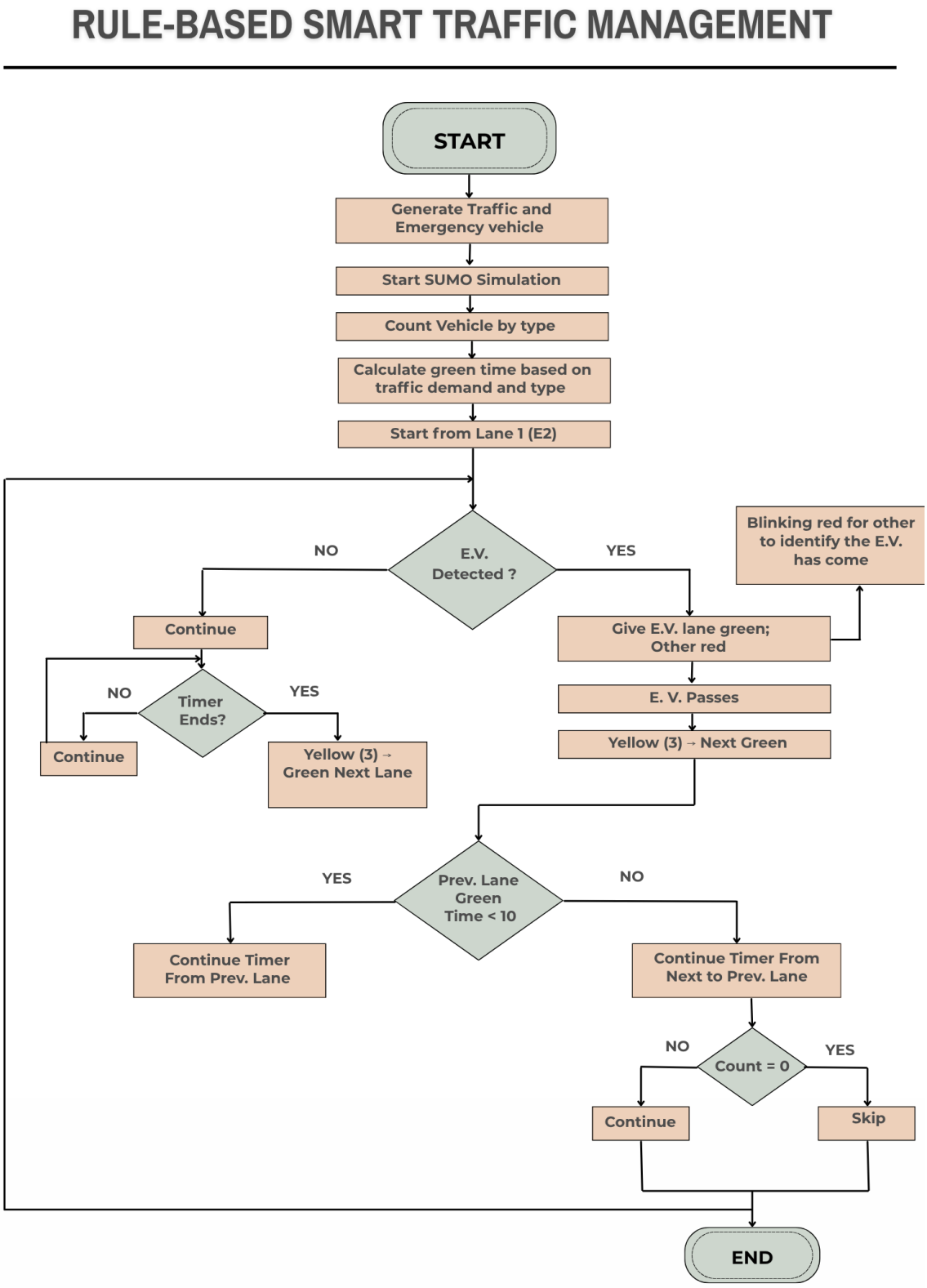


Diagram 3. All Rule Flow chart

Chapter 4

IMPLEMENTATION, TESTING & MAINTENANCE

4.1 Implementation Tools

The Junction IQ - Smart Traffic Management System for Urban Congestion Control is implemented using SUMO, NetEdit, Python, and the TraCI API.**SUMO (Simulation of Urban Mobility):** SUMO is an open-source traffic simulator used to simulate individual vehicles, their movements, and traffic flow at intersections. It allows the system to test different traffic conditions, vehicle types, and signal control strategies in a virtual environment. **NetEdit:** NetEdit is used to design and modify the road network. It helps create roads, lanes, junctions, and traffic signal configurations required for the simulation. **Python:** Python is used to develop the rule-based decision engine. It processes traffic information, including vehicle count, vehicle speed, to determine suitable green-light durations. It also implements the rules for emergency vehicle prioritization.**TraCI API:** The Traffic Control Interface (TraCI) connects Python with SUMO. It allows Python to retrieve vehicle information and send traffic commands while the simulation.

4.2 Coding Standards

The project follows basic coding standards to maintain code readability and reliability. Meaningful variable names, proper indentation, comments, and modular programming are used. The Python code is divided into functions for vehicle data collection, signal timing calculation, and emergency vehicle prioritization. Error handling helps manage connection and simulation issues.

4.3 Project Scheduling

Project scheduling involves dividing the development process into phases and allocating sufficient time to complete each activity. The project follows a systematic approach, beginning with requirement analysis and ending with testing, performance evaluation, and documentation. Initially, the problem statement is finalized, existing traffic management approaches are studied, and the system architecture is prepared. Next, SUMO is installed and the road network is created using NetEdit. After configuring the simulation environment, Python is connected to SUMO through the TraCI API.The next phase involves implementing the rule-based traffic control mechanism. Vehicle

count, lane density, speed, and waiting time are analyzed to determine suitable signal timings. Emergency vehicle prioritization is then implemented and integrated into the decision-making process.Finally, the complete system is tested under different simulated traffic conditions. The results of fixed-time and dynamic-time signal control are compared to evaluate the reduction in average waiting time. Necessary corrections are made, and the final report, diagrams, results, and presentation are prepared.

4.4 Testing

Testing verifies that the system works correctly under different simulated traffic conditions. Individual modules are checked for accurate vehicle data collection and signal timing decisions. Integration testing verifies communication between Python and SUMO, while functional testing checks dynamic signal control and ambulance prioritization.Performance testing compares fixed-time and dynamic-time signal control. The recorded results show that average waiting time decreased from **118 seconds to 70.48 seconds**, representing a **40.27% reduction** under the tested simulation conditions.

Observation Table :

Sr. No	Case	Total vehicle (count)	Car (%)	Bike (%)	Truck (%)	Bus (%)	Van (%)	Time with logic (Sec)	Optimization (%)
1	Less congestion, No E.V.	200	55	25	10	5	5	30	74.58
2	Less congestion, 1 E.V.	200	55	25	10	5	5	50	57.63
3	Less congestion, 2 E.V.	200	55	25	10	5	5	56	52.54
4	Less congestion, 2 E.V. in same lane	200	55	25	10	5	5	58	50.85
5	More congestion, No E.V	300	55	25	10	5	5	55	53.39
6	More congestion, 1 E.V. Green Time > 10 seconds for the previous lane	300	55	25	10	5	5	84	28.81
7	More congestion, 1 E.V. Green Time < 10 seconds for the previous lane	300	55	25	10	5	5	93	21.19
8	More congestion, 2 E.V.	300	55	25	10	5	5	104	11.86
9	More congestion, 2 E.V. in same lane	300	55	25	10	5	5	96	18.64
10	Bike count more, without E.V.	300	25	70	5	5	5	50	57.63
11	Bus count more, without E.V.	300	25	10	5	55	5	57	51.69
12	Truck count more, without E.V.	300	25	10	55	5	5	78	33.90
13	E.V. count more	300	55	25	10	5	5	165	-39.83
14	One lane is empty, and the other lanes are congested	300	55	25	10	5	5	44	62.71
15	Only one lane has vehicles.	300	55	25	10	5	5	13	88.98
16	Less congestion, 1 E.V. Green Time > 10 seconds for the previous lane	300	55	25	10	5	5	55	53.39
17	Less congestion, 1 E.V. Green Time < 10 seconds for the previous lane.	300	55	25	10	5	5	58	50.85
18	Bike count more, with 1 E.V.	300	25	70	5	5	5	60	49.15
19	Bus count more, with 1 E.V.	300	25	10	5	55	5	84	28.81
20	Truck count more, with 1 E.V.	300	25	10	55	5	5	100	15.25
21	2 E.V. at same time (1 Fire truck, 1 Ambulance)	300	55	25	10	5	5	76	35.59
22	2 E.V. at same time (2 Ambulance)	300	55	25	10	5	5	75	36.44
23	2 E.V. at same time (2 Fire truck)	300	55	25	10	5	5	70	40.68
24	4 E.V. at same time (2 Ambulance, 2 Fire truck)	300	35	25	10	5	5	75	36.44
25	4 E.V. at same time (4 Ambulance)	300	55	25	10	5	5	76	35.59
	AVERAGE	-	-	-	-	-	-	**70.48**	**40.27**

Table 5. Observation table with case and count of vehicles

Fixed Signal Timer : 118

Result	Time with logic	optimization (%)
Case - 1	30	74.57
Case - 2	50	57.62
Case - 3	56	52.54
Case - 4	58	50.84
Case - 5	55	53.38
Case - 6	84	28.81
Case - 7	93	21.18
Case - 8	104	11.86
Case - 9	96	18.64
Case - 10	50	57.62
Case - 11	57	51.69
Case - 12	78	33.89
Case - 13	165	-39.83
Case - 14	44	62.71
Case - 15	13	89.98
Case - 16	55	53.38
Case - 17	58	50.84
Case - 18	60	49.15
Case - 19	84	28.81
Case - 20	100	15.25
Case - 21	76	35.59
Case – 22	75	36.44
Case – 23	70	40.67
Case – 24	75	36.44
Case – 25	76	35.59
	Avg = 71.48	Avg = 40.27

Table 6. Observation table with case and respective times

$$\textbf{Optimization (\%)} = \frac{\textbf{Time without logic} - \textbf{Time with logic}}{\textbf{Time without logic}} \times \mathbf{100}$$

Chapter 5

RESULTS & DISCUSSIONS

5.1 User Interface Representation

The user interface of the Smart Traffic Management System is represented through the SUMO graphical user interface (GUI). It displays the road network, lanes, vehicles, and traffic signals during simulation. Users can observe vehicle movement, traffic congestion, and signal changes under different traffic conditions.The interface also helps visualize the behaviour of the dynamic signal control mechanism and ambulance prioritization. This visual representation makes it easier to understand and evaluate the system's performance.

5.2 Module Description

The project consists of several modules that work together to manage traffic signals dynamically. The **Road Network Design Module** uses NetEdit to create roads, lanes, and intersections.The **Traffic Simulation Module** uses SUMO to simulate vehicle movement and traffic flow. The **Data Collection Module** collects vehicle count, lane density, speed, and waiting time through TraCI.
The **Decision-Making Module**, implemented in Python, analyzes traffic conditions and determines suitable green-light durations using predefined rules.The **Emergency Vehicle Prioritization Module** detects ambulances in the simulation and applies the configured priority rules. Finally, the **Signal Control Module** sends commands through TraCI to update traffic signals according to the calculated timings

5.3 Snapshots of System

System snapshots provide visual evidence of the working of the Smart Traffic Management System. Screenshots of the SUMO simulation can show the designed road network, vehicle movement, traffic signals, and congestion under different traffic conditions.Additional snapshots can demonstrate dynamic signal timing and ambulance prioritization. Screenshots of different test scenarios can also illustrate how the system responds to varying vehicle counts, heavy traffic, and emergency situations.These snapshots help demonstrate the implementation and functioning of the proposed system.

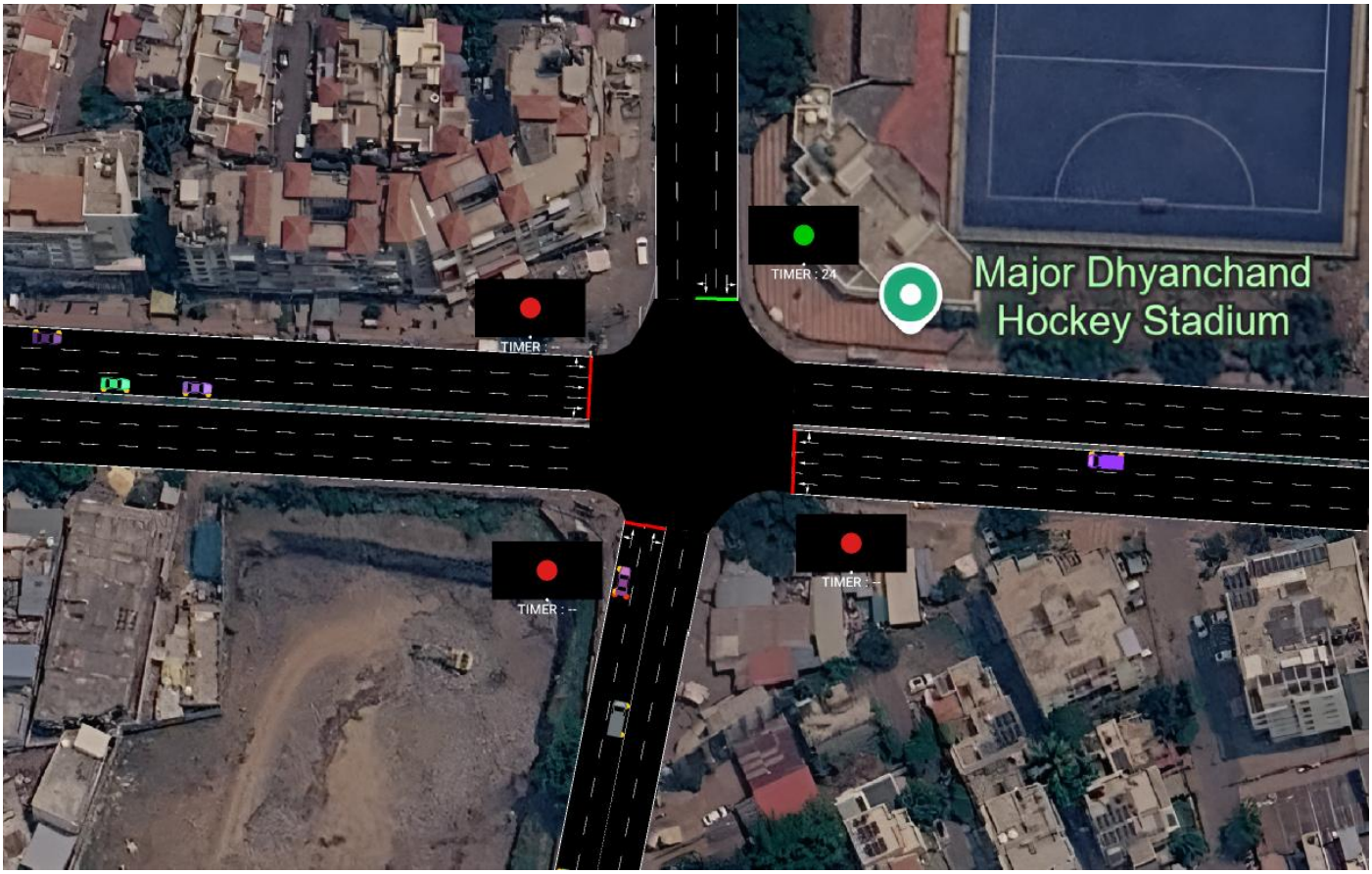


Diagram 4. Simulation without logic

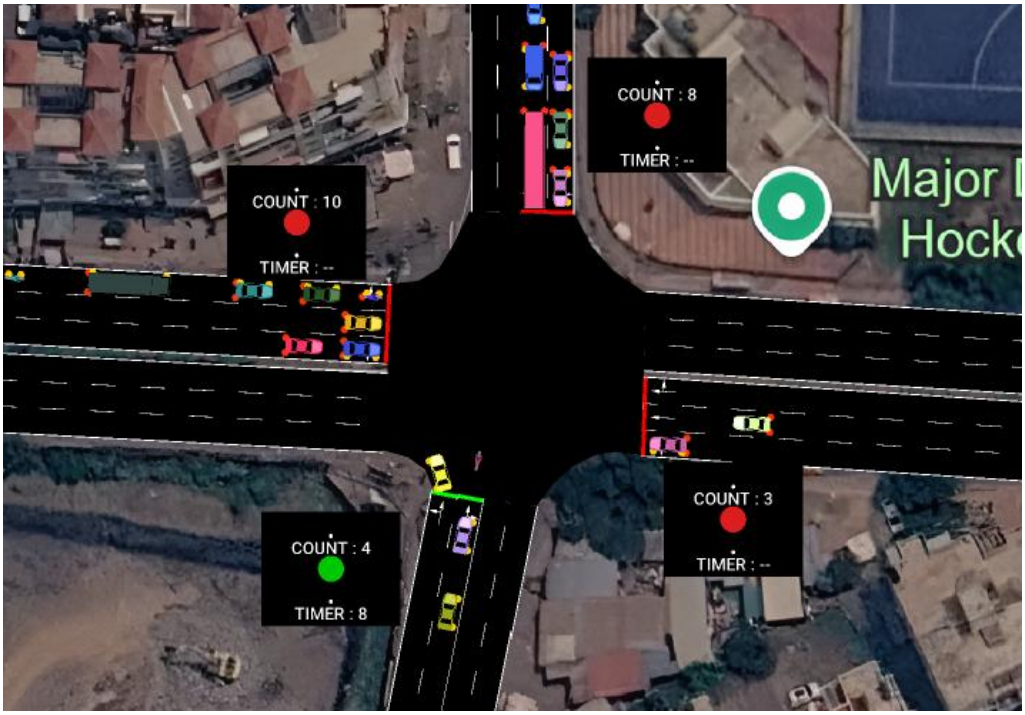


Diagram 5. Simulation with logic

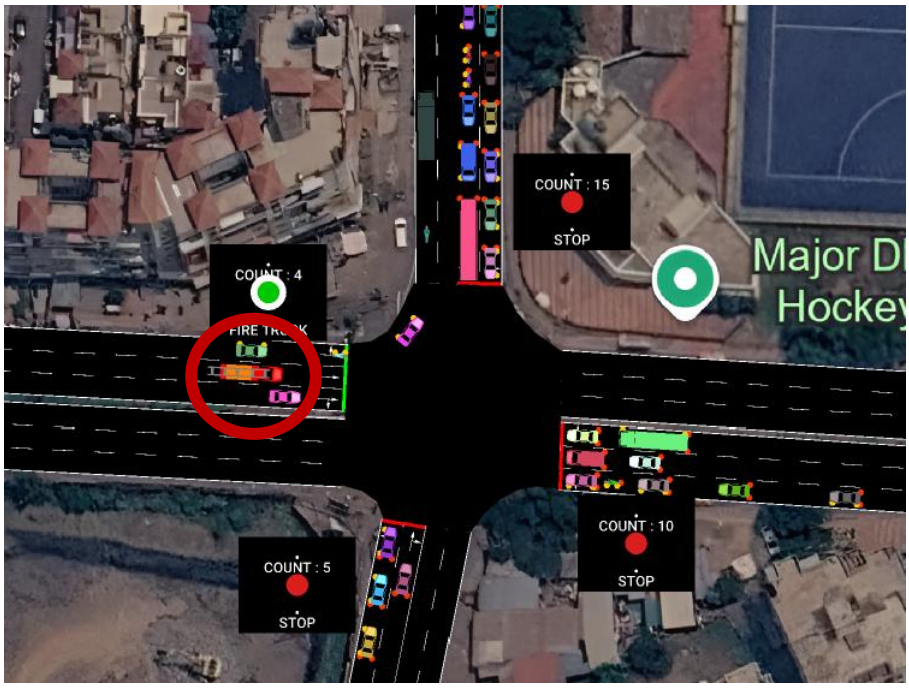


Diagram 6. E. V. Fire Truck



Diagram 7. E. V. Ambulance

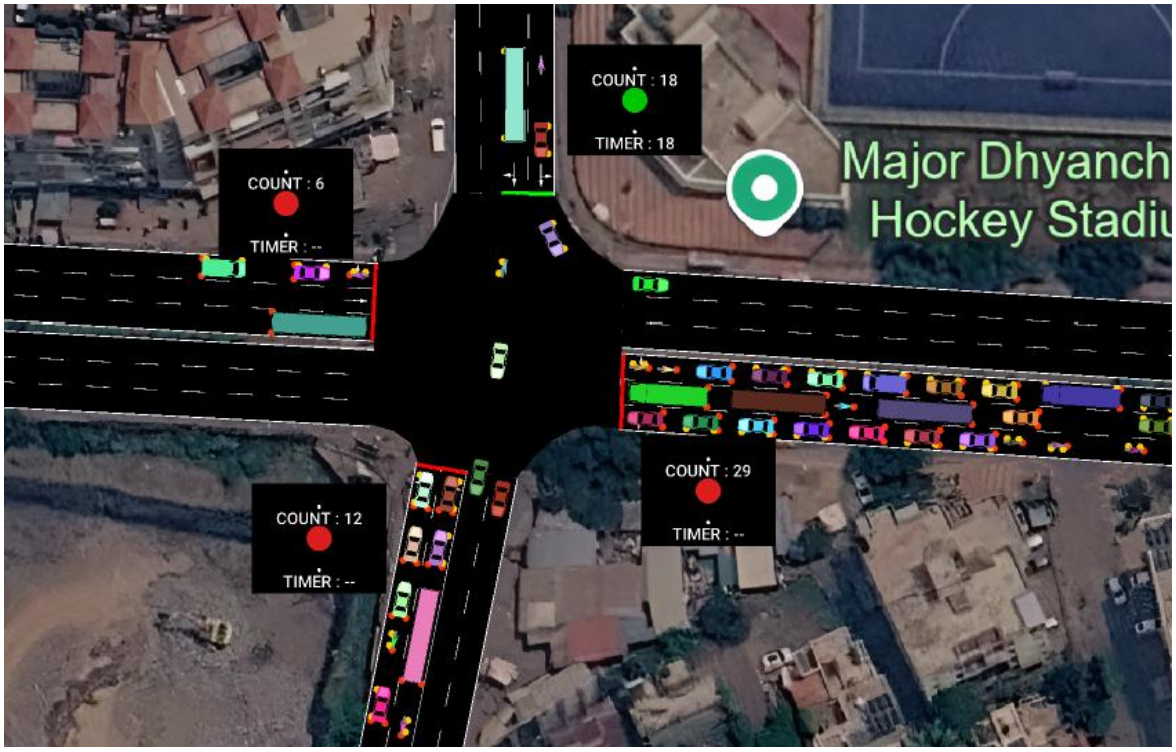


Diagram 8. Simulation with heavy congestion

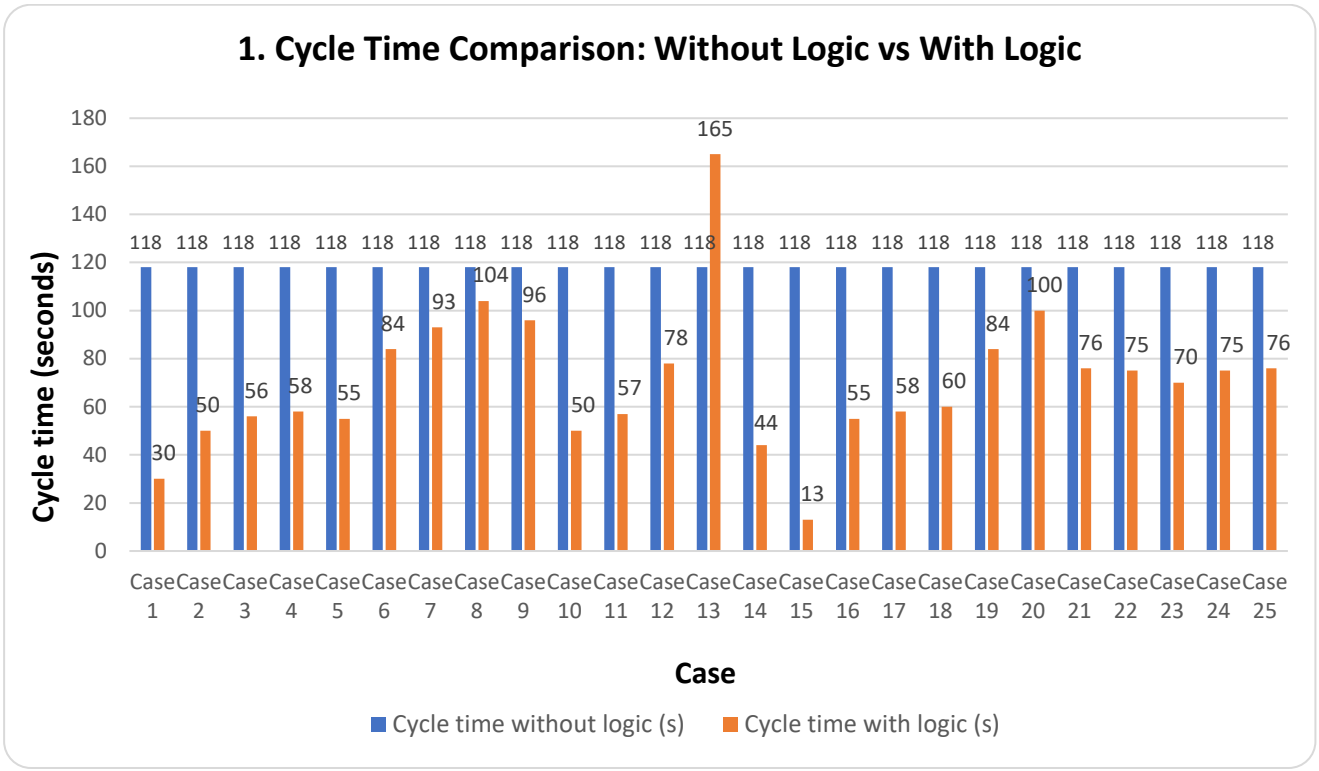


Diagram 9. Cycle Time Comparison — Without Logic vs With Logic

1. Graph 1: Cycle Time Comparison — Without Logic vs With Logic

The first graph compares the traffic signal cycle completion time without logic and with the proposed Smart Traffic Management System across 25 different cases. Without logic, the cycle time remains constant at 118 seconds for every case. However, with the proposed logic, the cycle time varies according to traffic congestion, vehicle types, and emergency vehicle conditions. The system achieves the best result in Case 15, where only one lane has vehicles, reducing the cycle time to 13 seconds. In contrast, Case 13 takes 165 seconds due to a higher E.V. count. The graph demonstrates that the proposed system reduces cycle time in most scenarios, although certain complex traffic conditions require further optimization.

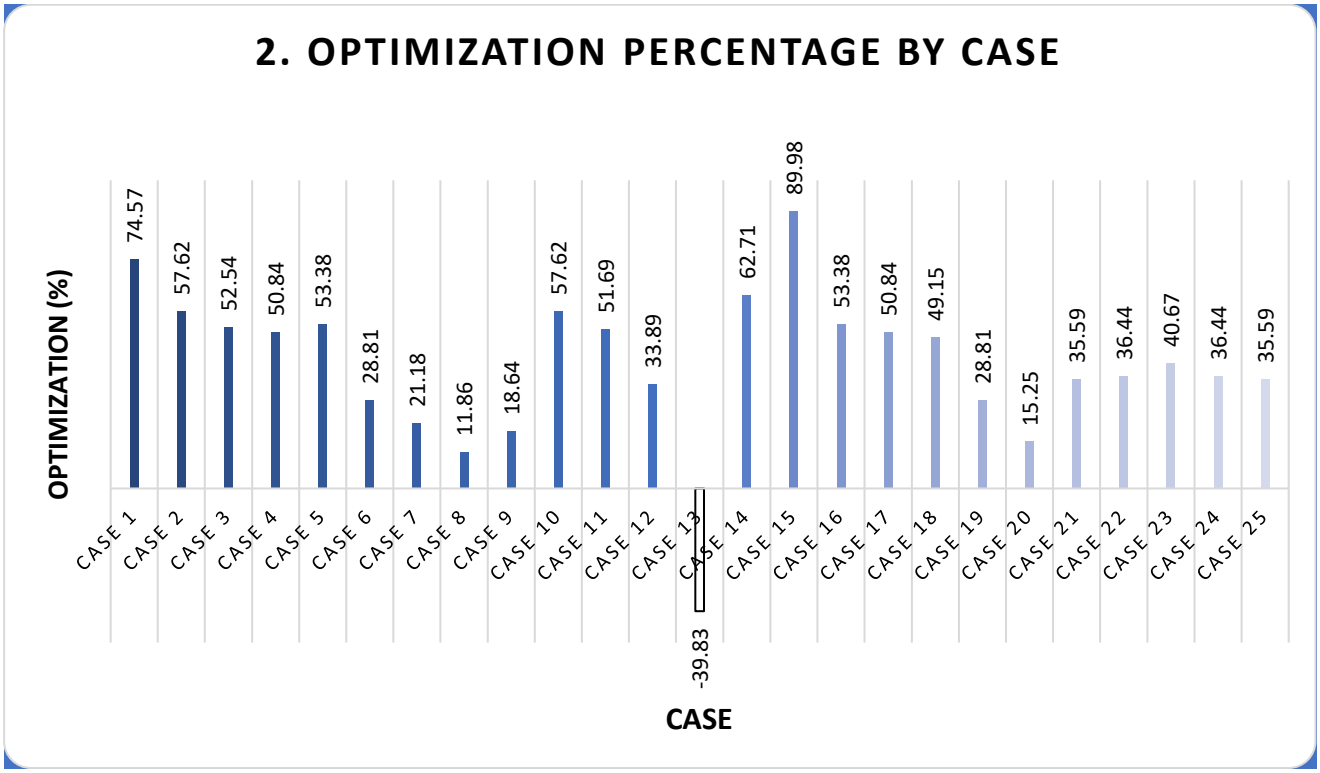


Diagram 10. Optimization Percentage by Case

2. Graph 2: Optimization Percentage by Case

The second graph represents the percentage of optimization achieved in each of the 20 cases by comparing the cycle time with and without the proposed logic. Positive values indicate a reduction in cycle time, whereas negative values indicate that the proposed system takes longer than the fixed-time system. The highest optimization is achieved in Case 15 at 89.98%, followed by Case 14 at 65.07% and Case 1 at 61.9%. However, Case 20 shows the lowest optimization at –26.98%, followed by Case 12 at –22.22%, indicating increased cycle completion times in these scenarios. Overall, the system achieves an average optimization of approximately 31.75%, demonstrating improved traffic signal efficiency in most tested conditions.

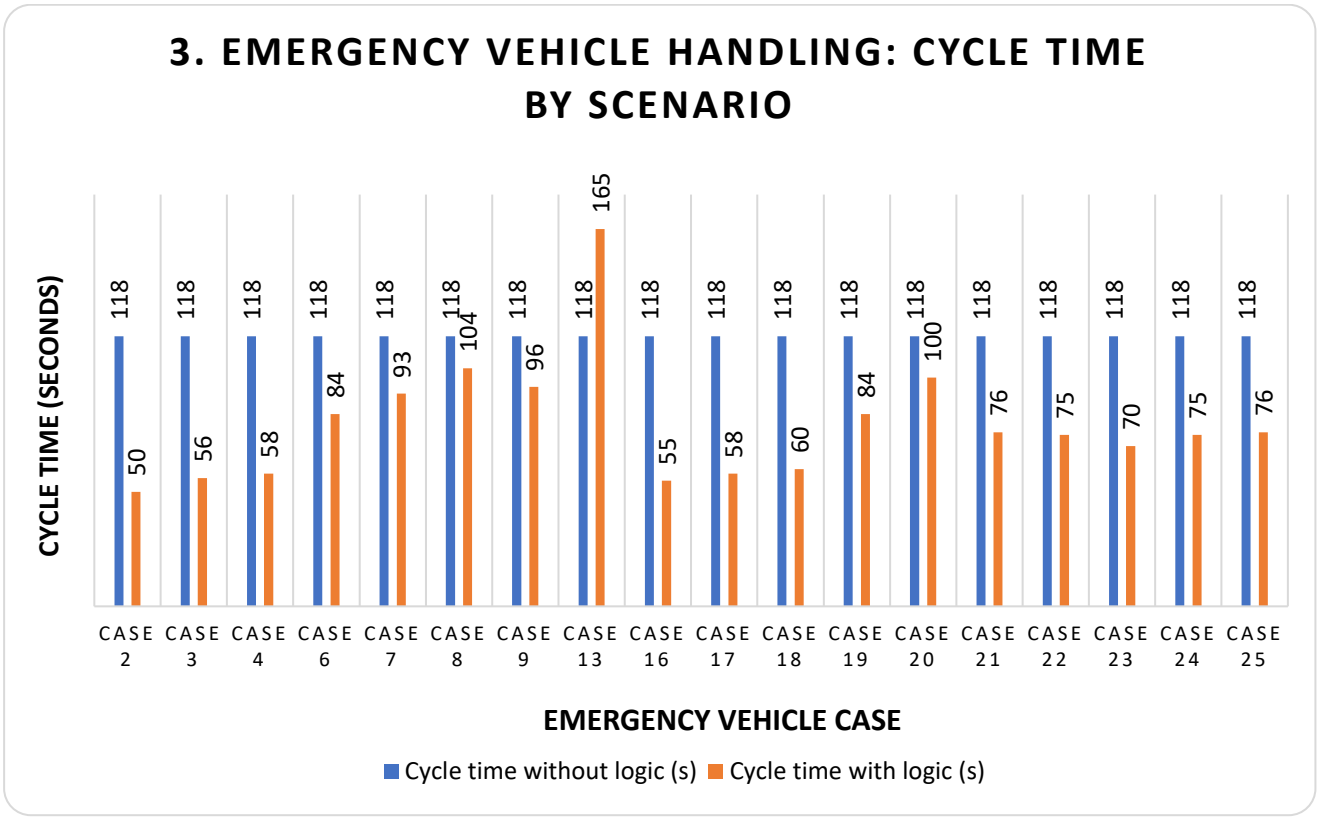


Diagram 11. Emergency Vehicle Handling — Cycle Time by Scenario

3. Graph 3: Emergency Vehicle Handling — Cycle Time by Scenario

The third graph illustrates the cycle completion time for 13 scenarios involving emergency vehicle handling. The fixed-time system maintains a constant cycle time of 126 seconds, while the proposed system adjusts the timing according to different traffic and emergency vehicle conditions. In several scenarios, the proposed logic reduces the cycle time, such as Scenario 1, where it decreases to 65 seconds, and Scenario 9, where it decreases to 60 seconds. However, in Scenario 5, the cycle time increases to 132 seconds, while Scenario 13 reaches 160 seconds, indicating challenges in certain traffic conditions involving emergency vehicles. These results highlight the importance of emergency vehicle prioritization while also showing that the current logic needs further improvement to handle complex scenarios efficiently.

5.4 Back-End Representation

The back end of the system is developed using Python and the TraCI API. Python processes traffic information received from SUMO and applies rule-based logic to determine signal timings. The system analyzes vehicle count, lane density, speed, and waiting time to identify traffic demand. It also checks for ambulance presence and applies the predefined emergency-priority mechanism.Through TraCI, Python sends traffic signal control commands to SUMO, which updates the signal states and simulates vehicle movement. This process repeats continuously until the simulation ends, allowing the system's performance to be evaluated under different traffic scenarios.

Chapter 6

CONCLUSION & FUTURE SCOPES

6.1 Conclusion

The Smart Traffic Management System for Urban Congestion Control provides a simulation-based approach to improving traffic signal management. The system uses SUMO, NetEdit, Python, and the TraCI API to analyze vehicle count, lane density, speed, and waiting time to dynamically adjust traffic signal timings. It also incorporates a rule-based mechanism to prioritize ambulances under predefined conditions.The system was evaluated using 25 different traffic scenarios, including varying congestion levels, different vehicle types, and emergency vehicle situations.The observation results show that the average time decreased from 118 seconds without the decision-making logic to 70.48 seconds with the logic, achieving a reported average optimization of **40.27%**. However, some scenarios showed increased time, indicating that further improvements are needed. Overall, the project demonstrates the potential of dynamic traffic signal control to reduce unnecessary waiting and improve traffic management in a simulated environment.

6.2 Future Scope

The project can be enhanced in several ways to improve its performance and practical applicability. In the future, real-time traffic data from cameras, sensors, or IoT devices could be integrated to make signal decisions based on actual road conditions. Machine learning techniques could be explored to improve traffic prediction and signal timing decisions. The system could also be extended to coordinate multiple intersections to reduce congestion across larger road networks. Additionally, the emergency vehicle prioritization mechanism can be improved to handle multiple ambulances and different emergency situations more efficiently. Further optimization of the rules for heavy vehicles, such as buses and trucks, could help address scenarios where waiting time increases. Finally, the system could be tested on more complex road networks and evaluated for potential real-world deployment.

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